2024 AMC 10B Problem 6 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10B Problem 6. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10B Problem 6

Problem:

A rectangle has integer length sides and an area of 2024 . What is the least possible perimeter of the rectangle?

Answer Choices:

A. 160

B. 180

C. 222

D. 228

E. 390

Solution:

Note that $2024 = 44 \cdot 46$. A 44×46 rectangle will have perimeter $2(44+46) = 180$. It is straightforward to check the other possible dimensions to show that this gives the rectangle with the least possible perimeter:

- 23×88 gives a perimeter of $2(23+88) = 222$.
- 22×92 gives a perimeter of $2(22+92) = 228$.
- 11×184 gives a perimeter of $2(11+184) = 390$.
- If one of the dimensions is $1, 2, 4,$ or 8 , then the other dimension is greater than 200 , yielding rectangles with greater perimeters.

Thus the least possible perimeter is $\text{(B)}\boxed{180}$.